computer networks

Credit Hours: 2-1

Course Outline

Introduction to Computer Networks, Network Models and Topologies, The OSI and TCP/IP Reference Models, Data Transmission Concepts and Techniques, Transmission Media, Data Link Layer and Protocols, Error Detection and Correction, Flow and Error Control Protocols, Medium Access Control (MAC) Techniques, Local Area Networks (LANs) and Ethernet, Network Layer and IP Addressing, Routing Algorithms and Protocols, Transport Layer Protocols (TCP and UDP), Congestion Control, Application Layer Protocols (HTTP, DNS, FTP, SMTP), Network Security Fundamentals, Wireless and Mobile Networks, Emerging Networking Technologies (IoT, Cloud Networking, 5G

16-Week Course Plan

Week 1

Week 1: Introduction to Computer Networks

Lecture 1.1: Introduction to Computer Networks (Theory)

Learning Objectives

* Define computer networks and their importance

* Identify types of computer networks

Concepts

Computer networks are collections of interconnected devices that communicate with each other. They are essential for sharing resources, facilitating communication, and enabling global connectivity. There are two primary types of computer networks: Local Area Networks (LANs) and Wide Area Networks (WANs). LANs connect devices within a limited geographical area, while WANs span larger distances.

Lecture 1.2: Network Models and Topologies (Theory)

Learning Objectives

* Explain the concept of network models

* Describe different network topologies

Concepts

Network models provide a framework for understanding and designing computer networks. The OSI and TCP/IP reference models are the two primary frameworks used today. Network topologies refer to the physical and logical arrangement of devices in a network. Common network topologies include Bus, Star, Ring, and Mesh.

Lecture 1.3: Course Overview and Expectations (Theory)

Learning Objectives

* Introduce the course outline and schedule

* Set expectations for the course

Concepts

This course will cover the fundamental concepts of computer networks, including network models, topologies, protocols, and security. The course will consist of 16 weeks, with 2-1 credit hours. The lecture portion will cover theoretical concepts, while the lab will provide hands-on experience with network simulation tools.

Lab 1: Getting Familiar with Network Simulation Tools

Learning Objectives

* Familiarize students with network simulation tools

* Set up a basic network simulation environment

Concepts

In this lab, students will be introduced to network simulation tools, such as GNS3 and Cisco Packet Tracer. They will learn how to set up a basic network simulation environment, including configuring routers and switches, and verifying network connectivity.

Week 2

Week 2: Network Models and Topologies

Lecture 1: Introduction to Network Models (Theory, 30 minutes)

TCP/IP Reference Model Overview

The TCP/IP Reference Model is a 4-layered model consisting of:

• Application Layer

• Transport Layer

• Internet Layer

• Network Access Layer

Learning Objectives:

• Understand the TCP/IP Reference Model

• Identify the four layers of the TCP/IP Model

Concepts:

• Network models are conceptual frameworks for organizing network architecture

• TCP/IP Model is a commonly used reference model in computer networking

Lecture 2: OSI Reference Model (Theory, 30 minutes)

OSI Reference Model Overview

The OSI Reference Model is a 7-layered model consisting of:

• Physical Layer

• Data Link Layer

• Network Layer

• Transport Layer

• Session Layer

• Presentation Layer

• Application Layer

Learning Objectives:

• Understand the OSI Reference Model

• Identify the seven layers of the OSI Model

Concepts:

• OSI Model provides a comprehensive framework for understanding network architecture

• Each layer has a specific function in the communication process

Lecture 3: Network Topologies (Theory, 30 minutes)

Network Topologies Overview

Network topologies refer to the physical and logical arrangement of devices in a network.

• Bus Topology

• Star Topology

• Ring Topology

• Mesh Topology

Learning Objectives:

• Understand different types of network topologies

• Identify the advantages and disadvantages of each topology

Concepts:

• Network topology affects network performance, reliability, and scalability

Lecture 4: Network Topology Advantages and Disadvantages (Theory, 30 minutes)

Advantages and Disadvantages of Network Topologies

Each network topology has its advantages and disadvantages.

• Bus Topology: easy to install, but prone to network failures

• Star Topology: easy to troubleshoot, but expensive

• Ring Topology: efficient, but prone to network failures

• Mesh Topology: reliable, but expensive to install and maintain

Learning Objectives:

• Identify the advantages and disadvantages of each network topology

Concepts:

• Understanding network topology is crucial for network design and implementation

Lab: Network Topology Simulation (Lab, 60 minutes)

* Set up a network topology simulation using a network simulator or emulator

* Experiment with different network topologies and observe their performance

* Analyze the advantages and disadvantages of each topology

Learning Objectives:

• Understand the practical implementation of network topologies

• Compare the performance of different network topologies

Week 3

Week 3: Data Transmission Concepts and Techniques

Lecture 1: Introduction to Data Transmission (Theory)

Data Transmission Fundamentals

Data transmission is the process of sending and receiving data over a communication network. It involves the conversion of data into a signal that can be transmitted over a medium, such as a wire or through the air.

Learning Objectives

* Define data transmission and its importance in computer networks

* Explain the different types of data transmission

Concepts

* Data transmission is a critical component of computer networks

* Data transmission involves signal conversion, modulation, and demodulation

* Types of data transmission include:

+ Analog transmission

+ Digital transmission

+ Hybrid transmission

Lecture 2: Data Encoding and Modulation (Theory)

Data Encoding and Modulation

Data encoding and modulation are crucial steps in the data transmission process. Encoding converts data into a signal, while modulation changes the signal's characteristics to transmit it over a medium.

Learning Objectives

* Explain the purpose of data encoding and modulation

* Describe the different types of data encoding and modulation techniques

Concepts

* Data encoding:

+ Binary encoding

+ ASCII encoding

+ Unicode encoding

* Data modulation:

+ Amplitude shift keying (ASK)

+ Frequency shift keying (FSK)

+ Phase shift keying (PSK)

Lecture 3: Bit Rate and Bandwidth (Theory)

Bit Rate and Bandwidth

Bit rate and bandwidth are essential concepts in data transmission. Bit rate refers to the number of bits transmitted per second, while bandwidth refers to the range of frequencies available for data transmission.

Learning Objectives

* Define bit rate and bandwidth

* Explain the relationship between bit rate and bandwidth

Concepts

* Bit rate:

+ Measured in bits per second (bps)

+ Affects the speed of data transmission

* Bandwidth:

+ Measured in hertz (Hz)

+ Determines the range of frequencies available for data transmission

Lab: Measuring Bit Rate and Bandwidth (Lab)

Measuring Bit Rate and Bandwidth

In this lab, you will measure the bit rate and bandwidth of a given data transmission system using various tools and equipment.

Learning Objectives

* Measure the bit rate and bandwidth of a data transmission system

* Analyze the relationship between bit rate and bandwidth

Tasks

* Measure the bit rate of a data transmission system using a bit rate meter

* Measure the bandwidth of a data transmission system using a spectrum analyzer

* Analyze the data collected and draw conclusions about the relationship between bit rate and bandwidth

Week 4

Week 4: Network Layer and IP Addressing

Lecture 1 (Theory - 1.5 hours): Introduction to Network Layer (45 minutes) and IP Addressing (45 minutes)

Network Layer

Learning Objectives:

• Understand the role of the Network Layer in the OSI model

• Identify the key functions of the Network Layer

Concepts:

The Network Layer is responsible for routing data between different networks. It provides logical addressing and routing functionality. The Network Layer uses logical addresses, such as IP addresses, to identify devices on a network.

IP Addressing

Learning Objectives:

• Understand the basics of IP addressing

• Identify the different types of IP addresses

Concepts:

IP addresses are 32-bit numbers used to identify devices on a network. There are two types of IP addresses: IPv4 and IPv6. IPv4 uses dotted decimal notation, while IPv6 uses hexadecimal notation. IP addresses are divided into network and host parts.

Lecture 2 (Theory - 1.5 hours): Subnetting and CIDR

Subnetting

Learning Objectives:

• Understand the concept of subnetting

• Identify the benefits of subnetting

Concepts:

Subnetting is a technique used to divide a network into smaller sub networks. It involves dividing the host part of an IP address into sub network and host parts. Subnetting helps to conserve IP addresses and improve network organization.

CIDR

Learning Objectives:

• Understand the concept of CIDR

• Identify the benefits of using CIDR

Concepts:

CIDR (Classless Inter-Domain Routing) is a method of representing IP addresses in a more compact form. It involves using a prefix length to specify the number of bits in the network part of an IP address. CIDR helps to reduce the size of routing tables and improve network scalability.

Lecture 3 (Theory - 1.5 hours): Routing Algorithms and Protocols

Routing Algorithms

Learning Objectives:

• Understand the different types of routing algorithms

• Identify the benefits of each routing algorithm

Concepts:

There are two main types of routing algorithms: distance-vector and link-state. Distance-vector algorithms, such as RIP, use the distance to a network to determine the best route. Link-state algorithms, such as OSPF, use the state of a network to determine the best route.

Routing Protocols

Learning Objectives:

• Understand the different types of routing protocols

• Identify the benefits of each routing protocol

Concepts:

There are two main types of routing protocols: interior gateway protocols (IGPs) and exterior gateway protocols (EGPs). IGPs, such as OSPF and RIP, are used within an autonomous system. EGPs, such as BGP, are used between autonomous systems.

Lab (1 hour): Configure IP Addressing and Subnetting on a Network Simulator

In this lab, students will configure IP addressing and subnetting on a network simulator. They will learn how to divide a network into sub networks and assign IP addresses to devices on each subnet.

Week 5

Week 5

Credit Hours: 2-1 (2 theory, 1 lab)

Lecture 1: Network Layer and IP Addressing (Theory, 45 minutes)

Network Layer and IP Addressing

Learning Objectives:

• Understand the Network Layer and its functions

• Learn about IP addressing and its types

• Understand subnet masking and CIDR

Concepts:

The Network Layer is the third layer of the OSI model and is responsible for routing data between devices on a network. It uses logical addressing, which is independent of the physical network topology. The Network Layer also provides connectionless and connection-oriented services to upper-layer protocols.

IP addressing is a fundamental concept in computer networking. IP addresses are logical addresses assigned to devices on a network. There are two types of IP addresses: IPv4 and IPv6. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.

Subnet masking is a technique used to divide an IP network into smaller subnetworks. It involves applying a mask to the IP address to create a subnet ID. CIDR (Classless Inter-Domain Routing) is a method of allocating IP addresses and routing that replaces the traditional classful routing hierarchy with a more flexible and efficient system.

Lecture 2: Routing Algorithms and Protocols (Theory, 45 minutes)

Routing Algorithms and Protocols

Learning Objectives:

• Understand routing algorithms and protocols

• Learn about distance-vector and link-state routing

• Understand routing metrics and routing tables

Concepts:

Routing algorithms and protocols are used to determine the best path for data to travel between devices on a network. There are two main types of routing algorithms: distance-vector and link-state.

Distance-vector routing algorithms use a table to store routing information and update it based on the distance to each network. Link-state routing algorithms use a table to store information about the network topology and update it based on changes in the network.

Routing metrics are used to determine the best path for data to travel. Common routing metrics include hop count, delay, and bandwidth. Routing tables are used to store information about the best path for data to travel to each network.

Lecture 3: Routing Algorithms and Protocols (Theory, 45 minutes)

Routing Algorithms and Protocols (continued)

Learning Objectives:

• Understand routing protocols and their types

• Learn about interior and exterior routing protocols

• Understand BGP and its role in routing

Concepts:

Routing protocols are used to exchange routing information between devices on a network. There are two main types of routing protocols: interior and exterior.

Interior routing protocols, such as RIP, OSPF, and EIGRP, are used within an autonomous system (AS). Exterior routing protocols, such as BGP, are used between autonomous systems.

BGP (Border Gateway Protocol) is a path-vector routing protocol used to exchange routing information between autonomous systems. It uses a path-vector algorithm to determine the best path for data to travel.

Lab: Network Layer and IP Addressing (Lab, 1 hour)

Lab Description: In this lab, students will learn how to configure IP addresses on a device, subnet a network, and use CIDR to allocate IP addresses. They will also learn how to use subnets to divide a network into smaller subnetworks.

Lab Objectives:

• Configure IP addresses on a device

• Subnet a network using CIDR

• Use subnets to divide a network into smaller subnetworks

Week 6

Week 6: Network Layer and IP Addressing

Lecture 1: Network Layer Overview (Theory - 50 minutes)

Network Layer Overview

The Network Layer is the third layer of the OSI model, responsible for routing data between different networks. Its primary functions include:

* Logical addressing and routing

* Fragmentation and reassembly of data

* Error-reporting and flow control

Learning Objectives:

* Understand the functions of the Network Layer

* Identify the importance of logical addressing and routing

Concepts:

* Network Layer protocols (e.g. IP, ICMP, IGMP)

* Routing concepts (e.g. routing tables, forwarding)

Lecture 2: IP Addressing (Theory - 50 minutes)

IP Addressing

IP addresses are used to identify devices on a network. There are two main types of IP addresses:

* IPv4: 32-bit address space, divided into five classes (A, B, C, D, E)

* IPv6: 128-bit address space, designed to replace IPv4

Learning Objectives:

* Understand the difference between IPv4 and IPv6

* Identify the structure of IP addresses

Concepts:

* IP address classes (A, B, C, D, E)

* IP address notation (dotted decimal notation)

Lecture 3: Subnetting and CIDR (Theory - 50 minutes)

Subnetting and CIDR

Subnetting is the process of dividing an IP network into smaller sub-networks. CIDR (Classless Inter-Domain Routing) is a method of representing IP addresses and subnet masks.

Learning Objectives:

* Understand the importance of subnetting and CIDR

* Calculate subnet masks and addresses

Concepts:

* Subnetting concepts (e.g. subnet mask, broadcast address)

* CIDR notation (e.g. /24, /30)

Lab: Network Layer and IP Addressing (Lab - 60 minutes)

* Configure and troubleshoot IP addressing on a network

* Practice subnetting and CIDR calculations using online tools and software

Week 7

Week 7: Network Layer and IP Addressing, Routing Algorithms and Protocols

Lecture 1: Network Layer and IP Addressing (1.5 hours)

Learning Objectives:

* Understand the Network Layer's role in routing and addressing

* Learn about IP address formats and classes

* Understand the concept of subnetting

Concepts:

The Network Layer is responsible for routing data between devices on different networks. IP (Internet Protocol) addresses are used to identify devices on a network. There are five IP address classes: A, B, C, D, and E. Classes A, B, and C are used for unicast addressing, while Class D is used for multicast addressing. Class E is reserved for future use. Subnetting involves dividing a network into smaller sub-networks to improve routing efficiency.

Lecture 2: Routing Algorithms and Protocols (1 hour)

Learning Objectives:

* Understand the basics of routing algorithms

* Learn about distance-vector and link-state routing protocols

* Understand the concept of routing tables

Concepts:

Routing algorithms determine the best path for data to travel between networks. Distance-vector routing protocols, such as RIP (Routing Information Protocol), use hop counts to determine the best path. Link-state routing protocols, such as OSPF (Open Shortest Path First), use network topology information to determine the best path. Routing tables are used to store the best path information for each network.

Lecture 3: Routing Algorithms and Protocols (continued) (1 hour)

Learning Objectives:

* Understand the differences between RIP and OSPF

* Learn about other routing protocols, such as BGP (Border Gateway Protocol)

* Understand the concept of route summarization

Concepts:

RIP is a distance-vector protocol, while OSPF is a link-state protocol. BGP is a path-vector protocol used for inter-domain routing. Route summarization involves aggregating multiple routes into a single route.

Lab 1: Configuring IP Addresses and Routing Protocols (2 hours)

Learning Objectives:

* Configure IP addresses on a network

* Configure routing protocols, such as RIP and OSPF

* Understand how to troubleshoot routing issues

Concepts:

In this lab, students will configure IP addresses on a network and configure routing protocols using RIP and OSPF. Students will also learn how to troubleshoot routing issues.

Week 8

Week 8: Network Security Fundamentals

Lecture 1: Introduction to Network Security Fundamentals (1 hour)

Lecture Title: Network Security Fundamentals

Learning Objectives

• Understand the importance of network security

• Identify common network security threats

• Learn basic security measures

Concepts

Network security is crucial in ensuring the confidentiality, integrity, and availability of data transmitted over computer networks. With the increasing reliance on networks for communication and data transfer, network security has become a top priority. Common network security threats include hacking, malware, and unauthorized access. Basic security measures include firewalls, encryption, and secure authentication protocols.

Lecture 2: Network Threats and Vulnerabilities (1 hour)

Lecture Title: Network Threats and Vulnerabilities

Learning Objectives

• Identify types of network threats

• Understand network vulnerabilities

• Learn how to mitigate network threats

Concepts

Network threats can be categorized into two main types: external threats and internal threats. External threats include hacking, malware, and denial-of-service (DoS) attacks. Internal threats include unauthorized access and insider attacks. Network vulnerabilities include weak passwords, outdated software, and misconfigured firewalls. To mitigate network threats, firewalls, intrusion detection systems, and antivirus software can be used.

Lecture 3: Network Security Protocols (1 hour)

Lecture Title: Network Security Protocols

Learning Objectives

• Learn about network security protocols

• Understand the role of encryption in network security

• Learn about secure authentication protocols

Concepts

Network security protocols play a crucial role in ensuring the confidentiality and integrity of data transmitted over networks. Encryption is a common network security protocol that converts plaintext data into unreadable ciphertext. Secure authentication protocols, such as SSL/TLS and SSH, ensure that only authorized users can access network resources. These protocols are essential in preventing unauthorized access and maintaining network security.

Lab: Network Security Lab (2 hours)

Lab Title: Network Security Lab

Learning Objectives

• Configure a firewall to block unauthorized access

• Implement encryption to secure data transmission

• Configure secure authentication protocols

Concepts

In this lab, students will configure a firewall to block unauthorized access, implement encryption to secure data transmission, and configure secure authentication protocols. Students will also learn how to monitor network traffic and detect potential threats. The lab will provide hands-on experience with network security protocols and tools.

Week 9

Week 9: Network Layer and IP Addressing

Lecture 1: Introduction to Network Layer (Theory, 50 minutes)

Network Layer Overview

The network layer is the third layer of the OSI model, responsible for routing data between different networks. It provides logical addressing and routing services.

Learning Objectives

* Understand the network layer's functions

* Identify the network layer's protocols

Concepts

* Network layer protocols (e.g., IP, ICMP)

* Logical addressing (IP addresses)

* Routing tables

* Packet forwarding

Lecture 2: IP Addressing (Theory, 50 minutes)

IP Addressing Basics

IP addresses are used to identify devices on a network. They consist of a 32-bit or 128-bit address, divided into four or eight groups of numbers separated by dots.

Learning Objectives

* Understand IP address structure and notation

* Identify IP address classes (A, B, C, D, E)

Concepts

* IPv4 vs. IPv6

* Subnetting

* CIDR notation

Lecture 3: Subnetting and CIDR (Theory, 50 minutes)

Subnetting and CIDR Basics

Subnetting is the process of dividing a network into smaller sub-networks. CIDR (Classless Inter-Domain Routing) is a method of assigning addresses that allows for variable-length subnet masks.

Learning Objectives

* Understand subnetting and CIDR concepts

* Calculate subnet masks and addresses

Concepts

* Subnet mask calculation

* CIDR notation and prefix length

Lab: Network Layer and IP Addressing (Lab, 2 hours)

Lab Exercises

1. Configure a network with subnets and CIDR notation

2. Calculate subnet masks and addresses

3. Implement routing tables and packet forwarding

Note: The lab exercises will be provided in the lab manual.

Week 10

Week 10: Network Security Fundamentals

Lecture 1 (Theory): Network Security Fundamentals (≈150 words)

Network Security Fundamentals

Learning Objectives

• Understand the importance of network security

• Identify common network security threats

• Explain basic network security measures

Concepts

Network security is crucial to protect computer networks from unauthorized access, use, disclosure, disruption, modification, or destruction. Common network security threats include hacking, malware, phishing, and denial-of-service (DoS) attacks. Basic network security measures include firewalls, intrusion detection and prevention systems, encryption, and access control. Firewalls block unauthorized access to a network, while intrusion detection and prevention systems detect and prevent malicious activity. Encryption protects data in transit, and access control ensures that only authorized personnel can access network resources.

Lecture 2 (Theory): Network Security Protocols (≈150 words)

Network Security Protocols

Learning Objectives

• Understand network security protocols

• Explain the role of SSL/TLS in secure communication

• Identify other network security protocols

Concepts

Network security protocols are designed to secure communication over a network. Secure Sockets Layer/Transport Layer Security (SSL/TLS) is a widely used protocol for secure communication between a web browser and a web server. SSL/TLS encrypts data in transit, ensuring that sensitive information is protected from eavesdropping and tampering. Other network security protocols include Internet Protocol Security (IPsec), which provides secure communication between two endpoints, and Secure Shell (SSH), which provides secure access to a remote server.

Lecture 3 (Theory): Network Security Best Practices (≈150 words)

Network Security Best Practices

Learning Objectives

• Understand network security best practices

• Explain the importance of regular security audits

• Identify best practices for secure network configuration

Concepts

Network security best practices include regular security audits, secure network configuration, and employee education. Regular security audits help identify vulnerabilities and ensure compliance with security policies. Secure network configuration involves implementing firewalls, intrusion detection and prevention systems, and access control. Employee education is critical to prevent social engineering attacks, such as phishing and pretexting. By following these best practices, organizations can significantly reduce the risk of network security breaches and protect sensitive information.

Lab: Network Security Lab (≈150 words)

Network Security Lab

Learning Objectives

• Configure a firewall to block unauthorized access

• Implement intrusion detection and prevention systems

• Test secure communication using SSL/TLS

Concepts

In this lab, students will configure a firewall to block unauthorized access to a network, implement intrusion detection and prevention systems to detect and prevent malicious activity, and test secure communication using SSL/TLS. Students will also practice configuring secure network settings, such as enabling access control and encrypting data in transit. By completing this lab, students will gain hands-on experience with network security protocols and best practices.

Week 11

Week 11: Network Layer and IP Addressing

Lecture 1: Introduction to Network Layer (Theory, 1 hour)

Network Layer Overview

The network layer is responsible for routing data between devices on different networks. It provides logical addressing and routing functions.

Learning Objectives

* Understand the role of the network layer

* Identify the functions of the network layer

Concepts

* Network layer protocols (IP, ICMP, IGMP)

* Logical addressing (IP addresses)

* Routing tables and forwarding

Lecture 2: IP Addressing (Theory, 1 hour)

IP Addressing Basics

IP addresses are used to identify devices on a network. They consist of a network ID and a host ID.

Learning Objectives

* Understand the basics of IP addressing

* Identify the different types of IP addresses (IPv4, IPv6)

Concepts

* IP address structure (IPv4 and IPv6)

* Subnetting and supernetting

* Private and public IP addresses

Lecture 3: IP Address Classes and Subnetting (Theory, 1 hour)

IP Address Classes and Subnetting

IP addresses are divided into classes based on the first octet. Subnetting allows for more efficient use of IP addresses.

Learning Objectives

* Understand IP address classes

* Apply subnetting techniques

Concepts

* IP address classes (A, B, C)

* Subnet mask and subnet ID

* Calculating subnet masks and IDs

Lab: IP Address Configuration and Subnetting (Lab, 2 hours)

Lab Objectives

* Configure IP addresses on a network device

* Apply subnetting techniques to optimize IP address usage

Lab Tasks

* Configure a network device with a static IP address

* Apply subnetting to a network with multiple subnets

* Verify IP address configuration and subnetting using network utilities

Week 12

Week 12

Lecture 1: Network Layer and IP Addressing (Theory, 1 hour)

Network Layer and IP Addressing

Learning Objectives

• Understand the Network Layer's function in the OSI Model

• Learn about IP Addressing and its significance

• Understand the differences between IPv4 and IPv6

Concepts

The Network Layer is the third layer of the OSI Model, responsible for routing data between devices on a network. It provides logical addressing, routing, and congestion control. IP Addressing is a fundamental concept in networking, where each device on a network is assigned a unique IP address. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses. IPv6 was introduced to address the address exhaustion problem in IPv4.

Lecture 2: Network Layer and Routing Algorithms (Theory, 1 hour)

Network Layer and Routing Algorithms

Learning Objectives

• Understand various routing algorithms (Distance Vector, Link State, and Hybrid)

• Learn about routing tables and how they are updated

• Understand the differences between interior and exterior gateway protocols

Concepts

Routing algorithms are used to determine the best path for data to travel between devices on a network. Distance Vector algorithms (e.g., RIP) use hop count to determine the best path. Link State algorithms (e.g., OSPF) use network topology to determine the best path. Hybrid algorithms (e.g., EIGRP) combine aspects of both. Routing tables are updated based on routing protocols (IGPs and EGPs).

Lecture 3: Network Layer and IP Addressing (Theory, 1 hour)

Network Layer and IP Addressing

Learning Objectives

• Understand the significance of subnetting and CIDR

• Learn about variable length subnet masking (VLSM)

• Understand the concept of default gateways and how they are configured

Concepts

Subnetting is the process of dividing an IP address into multiple subnets. CIDR (Classless Inter-Domain Routing) is used to represent IP addresses in a more efficient manner. VLSM allows for variable length subnet masks. Default gateways are used to route traffic to devices that are not on the same subnet. They are configured on each device to point to the next-hop router.

Lab: Configuring Network Layer and IP Addressing (Lab, 2 hours)

Instructions: Configure a network with multiple subnets, using CIDR and VLSM. Set up default gateways on each device, and verify that traffic is routed correctly between devices on different subnets.

Week 13

Week 13: Network Security Fundamentals

Lecture 1: Introduction to Network Security (1 hour, 30 minutes)

Network Security Fundamentals

Learning Objectives

• Understand the importance of network security

• Identify common network security threats

• Explain the need for security protocols

Concepts

Network security is a critical aspect of computer networking. With the increasing reliance on networks, the potential for security breaches and attacks has grown. Network security is essential to protect against unauthorized access, data breaches, and other malicious activities. Common security threats include hacking, viruses, and malware. To address these threats, various security protocols and measures are implemented, such as firewalls, encryption, and access control.

Lecture 2: Types of Network Attacks (1 hour)

Types of Network Attacks

Learning Objectives

• Identify types of network attacks (intrusion, denial-of-service, man-in-the-middle)

• Explain the impact of each attack type

Concepts

Network attacks can be categorized into three main types: intrusion attacks, denial-of-service (DoS) attacks, and man-in-the-middle (MitM) attacks. Intrusion attacks involve unauthorized access or modification of data, while DoS attacks aim to overwhelm a network, rendering it unavailable. MitM attacks occur when an attacker intercepts and alters communication between two parties. Understanding these attack types is crucial for developing effective security strategies.

Lecture 3: Network Security Measures (1 hour)

Network Security Measures

Learning Objectives

• Explain the role of firewalls and encryption in network security

• Describe the purpose of access control and authentication

Concepts

Network security measures include firewalls, which control incoming and outgoing network traffic, and encryption, which protects data from interception. Access control and authentication ensure that only authorized users can access network resources. Firewalls can be hardware-based or software-based, and encryption methods include symmetric and asymmetric encryption. Understanding these measures is essential for designing and implementing secure networks.

Lab: Network Security Configuration and Testing (1 hour, 30 minutes)

Lab Instructions

• Set up a test network with a firewall and encryption

• Configure access control and authentication

• Test the network for vulnerabilities

Week 14

Week 14: Network Security Fundamentals

Lecture 1: Network Threats and Vulnerabilities (30 minutes, 1 hour of lecture time)

Network Threats and Vulnerabilities

Learning Objectives

• Understand various types of network threats and vulnerabilities

• Identify common network security risks

Concepts

Network threats and vulnerabilities are a critical aspect of network security. Network threats can be categorized into two main groups: external threats and internal threats. External threats come from outside the network, while internal threats originate from within the network. Common external threats include hacking, viruses, and denial-of-service (DoS) attacks. Internal threats include unauthorized access, impersonation, and data tampering.

Some common network security risks include:

• Unauthorized access to sensitive data

• Malware and virus attacks

• DoS and distributed denial-of-service (DDoS) attacks

• Eavesdropping and sniffing

• Man-in-the-middle (MitM) attacks

• SQL injection and cross-site scripting (XSS) attacks

Lecture 2: Network Security Measures (1 hour of lecture time)

Network Security Measures

Learning Objectives

• Understand various network security measures

• Identify common security protocols and technologies

Concepts

Network security measures are implemented to prevent, detect, and respond to network threats. Some common security measures include:

• Firewalls: Control incoming and outgoing network traffic based on predetermined security rules

• Encryption: Scramble data to prevent unauthorized access

• Authentication and authorization: Verify user identities and access levels

• Intrusion detection and prevention systems (IDPS): Monitor network traffic for signs of unauthorized access or malicious activity

• Virtual private networks (VPNs): Create secure, encrypted connections over public networks

• Secure socket layer (SSL)/transport layer security (TLS): Encrypt data in transit

Lecture 3: Network Security Protocols and Technologies (1 hour of lecture time)

Network Security Protocols and Technologies

Learning Objectives

• Understand various network security protocols and technologies

• Identify common security protocols and technologies

Concepts

Network security protocols and technologies are used to secure network communications. Some common security protocols and technologies include:

• Secure shell (SSH): Secure remote access to servers and devices

• Secure file transfer protocol (SFTP): Secure file transfers

• Virtual private networks (VPNs): Create secure, encrypted connections

• Multipurpose internet mail extensions (MIME): Secure email communications

• Public key infrastructure (PKI): Manage digital certificates and encryption keys

Lab Exercise: Network Security Measures (2 hours of lab time)

Lab Exercise: Network Security Measures

Objective

Configure and implement various network security measures to secure a network.

Tasks

• Configure a firewall to allow incoming traffic on specific ports

• Set up encryption using public key infrastructure (PKI)

• Implement an intrusion detection and prevention system (IDPS)

• Configure a virtual private network (VPN)

• Secure remote access using secure shell (SSH)

Week 15

Week 15: Application Layer Protocols

Lecture Title: Application Layer Protocols

Learning Objectives

• Understand the role of the Application Layer in the OSI model

• Identify and explain the functions of HTTP, DNS, FTP, and SMTP protocols

• Analyze the characteristics of each protocol

Lecture

The Application Layer is the 7th layer of the OSI model, responsible for providing services to end-user applications. It sits on top of the Transport Layer and provides services such as file transfer, email, and web browsing. The Application Layer is protocol-dependent, meaning it uses protocols specific to the service it provides.

HTTP (Hypertext Transfer Protocol)

* HTTP is a request-response protocol used for web browsing

* It uses TCP as its transport protocol

* HTTP has different versions, including HTTP/1.1 and HTTP/2

DNS (Domain Name System)

* DNS is a protocol that translates domain names to IP addresses

* It uses UDP as its transport protocol

* DNS caching and recursion improve efficiency

FTP (File Transfer Protocol)

* FTP is a protocol for transferring files over a network

* It uses TCP as its transport protocol

* FTP has two modes: active and passive

SMTP (Simple Mail Transfer Protocol)

* SMTP is a protocol for email transmission

* It uses TCP as its transport protocol

* SMTP uses a store-and-forward mechanism for message delivery

These protocols are essential components of the Internet, enabling communication between applications and services. Understanding their functions and characteristics is crucial for network administrators and developers.

Lecture Title: Application Layer Protocols (Continued)

Learning Objectives

• Explain the differences between HTTP, DNS, FTP, and SMTP

• Analyze the role of Application Layer protocols in network communication

Lecture

The Application Layer protocols discussed so far are all critical components of the Internet infrastructure. Understanding their differences and characteristics is essential for troubleshooting and developing network applications.

Comparison of HTTP, DNS, FTP, and SMTP

Protocol	Purpose	Transport Protocol	Key Features
HTTP	Web browsing	TCP	Request-response, caching
DNS	Domain name resolution	UDP	Caching, recursion, zone transfers
FTP	File transfer	TCP	Active and passive modes, file transfer modes
SMTP	Email transmission	TCP	Store-and-forward, email routing

Each protocol has its unique characteristics, and understanding these differences is crucial for network administration and development.

Lecture Title: Application Layer Protocols (Lab)

Learning Objectives

• Set up and configure HTTP, DNS, FTP, and SMTP protocols

• Analyze the behavior of each protocol in a simulated network environment

Lab

In this lab, you will set up and configure HTTP, DNS, FTP, and SMTP protocols using a virtual network simulator. You will analyze the behavior of each protocol in a simulated network environment, exploring their characteristics and interactions.

Lab Assignment:

1. Set up a web server using HTTP.

2. Configure a DNS server to resolve a domain name to an IP address.

3. Establish a file transfer using FTP.

4. Send an email using SMTP.

5. Analyze the behavior of each protocol in a simulated network environment.

This lab will provide hands-on experience with the Application Layer protocols, enabling you to understand their behavior and interactions in a network environment.

Week 16

Week 16: Emerging Networking Technologies

Lecture 1: Internet of Things (IoT) (Theory, 50 minutes)

Lecture Title: Introduction to IoT

Learning Objectives:

• Define the Internet of Things (IoT)

• Explain the benefits and challenges of IoT

• Identify the key components of IoT

Concepts:

The Internet of Things (IoT) refers to the network of physical devices, vehicles, home appliances, and other items embedded with sensors, software, and connectivity, allowing them to collect and exchange data with other devices and systems. The benefits of IoT include increased efficiency, improved decision-making, and enhanced user experience. However, IoT also poses challenges such as security risks, data management, and interoperability issues. The key components of IoT include sensors, actuators, gateways, and data analytics platforms.

Lecture 2: Cloud Networking (Theory, 50 minutes)

Lecture Title: Cloud Networking Fundamentals

Learning Objectives:

• Define cloud networking

• Explain the benefits and challenges of cloud networking

• Identify the key components of cloud networking

Concepts:

Cloud networking refers to the use of cloud computing and networking services to provide scalable and on-demand network resources. The benefits of cloud networking include reduced capital expenditures, increased flexibility, and improved disaster recovery. However, cloud networking also poses challenges such as security risks, data sovereignty, and latency issues. The key components of cloud networking include cloud service providers, virtual private clouds, and cloud-based security solutions.

Lecture 3: 5G Networks (Theory, 50 minutes)

Lecture Title: Introduction to 5G Networks

Learning Objectives:

• Define 5G networks

• Explain the benefits and challenges of 5G networks

• Identify the key features of 5G networks

Concepts:

5G networks are the fifth generation of wireless network technology, designed to provide faster data rates, lower latency, and greater connectivity than previous generations. The benefits of 5G networks include improved mobile broadband, enhanced IoT capabilities, and increased support for mission-critical communications. However, 5G networks also pose challenges such as security risks, spectrum allocation, and interoperability issues. The key features of 5G networks include massive machine-type communications, ultra-reliable low-latency communications, and enhanced mobile broadband.

Lab: Emerging Networking Technologies (1 hour)

• Set up an IoT device and connect it to the internet

• Configure a cloud networking platform and test its performance

• Demonstrate the features and benefits of 5G networks using a 5G-enabled device